EXPERIMENT 2: TITANIC DATASET ANALYSIS

AIM:

To analyze the Titanic dataset using Python by inspecting dataset information and summary statistics, handling missing values, visualizing passenger-class distribution, filtering passengers based on specified conditions, and splitting the data into training and testing sets.

ALGORITHM:

1. Import the required libraries: Pandas, Seaborn, Matplotlib, SimpleImputer, and train_test_split.

2. Load the Titanic CSV dataset into a Pandas DataFrame and display the first five rows.

3. Display the dataset shape, information, and summary statistics.

4. Fill missing values in the Age column using SimpleImputer with the mean strategy.

5. Replace missing values in the Cabin column with “Unknown”.

6. Fill missing values in the Embarked column using its mode.

7. Plot the passenger class distribution using a count plot.

8. Display female passengers who survived.

9. Display third-class passengers younger than 18 years.

10. Display first-class passengers older than 40 years.

11. Display surviving passengers in the specified category.

12. Display male passengers whose fare is greater than 100.

13. Display passengers who embarked from C and belonged to passenger class 2.

14. Display passengers with SibSp greater than 2.

15. Display non-surviving passengers who had no siblings/spouses and no parents/children aboard.

16. Display the five oldest surviving passengers.

17. Display passengers who paid zero fare.

18. Split the dataset into training and testing sets with test_size=0.2 and random_state=42.

```
# PRACTISE EXPERIMENT-1 Muthuraj(241801176)
# 1. Import Required Libraries
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder, StandardScaler
```

```
# PRACTISE EXPERIMENT-1 Muthuraj(241801176)
# 2. Load the Telco Customer Churn Dataset
df = pd.read_csv("C:\\Users\\ADMIN\\Downloads\\archive (1)\\Telco-Customer-Churn.csv")
print("Dataset Loaded Successfully!")
```

```
Dataset Loaded Successfully!
```

```
print("\nFirst 10 Records")
print(df.head(10))
```

```
First 10 Records
   customerID  gender  SeniorCitizen Partner Dependents  tenure PhoneService  \
0  7590-VHVEG  Female              0     Yes         No       1           No
1  5575-GNVDE    Male              0      No         No      34          Yes
2  3668-QPYBK    Male              0      No         No       2          Yes
3  7795-CFOCW    Male              0      No         No      45           No
4  9237-HQITU  Female              0      No         No       2          Yes
5  9305-CDSKC  Female              0      No         No       8          Yes
6  1452-KIOVK    Male              0      No        Yes      22          Yes
7  6713-OKOMC  Female              0      No         No      10           No
8  7892-POOKP  Female              0     Yes         No      28          Yes
9  6388-TABGU    Male              0      No        Yes      62          Yes

      MultipleLines InternetService OnlineSecurity  ... DeviceProtection  \
0  No phone service             DSL             No  ...               No
1                No             DSL            Yes  ...              Yes
2                No             DSL            Yes  ...               No
3  No phone service             DSL            Yes  ...              Yes
4                No     Fiber optic             No  ...               No
5               Yes     Fiber optic             No  ...              Yes
6               Yes     Fiber optic             No  ...               No
```

```
              PaymentMethod MonthlyCharges  TotalCharges Churn
0          Electronic check          29.85         29.85    No
1              Mailed check          56.95        1889.5    No
2              Mailed check          53.85        108.15   Yes
3  Bank transfer (automatic)          42.30       1840.75    No
4          Electronic check          70.70        151.65   Yes
5          Electronic check          99.65         820.5   Yes
6    Credit card (automatic)          89.10        1949.4    No
7              Mailed check          29.75         301.9    No
8          Electronic check         104.80       3046.05   Yes
9  Bank transfer (automatic)          56.15       3487.95    No

[10 rows x 21 columns]
```

```
print("\nLast 10 Records")
print(df.tail(10))
```

```
Last 10 Records
      customerID  gender  SeniorCitizen Partner Dependents  tenure  \
7033  9767-FFLEM    Male              0      No         No      38
7034  0639-TSIQW  Female              0      No         No      67
7035  8456-QDAVC    Male              0      No         No      19
7036  7750-EYXWZ  Female              0      No         No      12
7037  2569-WGERO  Female              0      No         No      72
7038  6840-RESVB    Male              0     Yes        Yes      24
7039  2234-XADUH  Female              0     Yes        Yes      72
7040  4801-JZAZL  Female              0     Yes        Yes      11
7041  8361-LTMKD    Male              1     Yes         No       4
7042  3186-AJIEK    Male              0      No         No      66

     PhoneService     MultipleLines InternetService       OnlineSecurity  ...  \
7033          Yes                No     Fiber optic                   No  ...
7034          Yes               Yes     Fiber optic                  Yes  ...
7035          Yes                No     Fiber optic                   No  ...
7036           No  No phone service             DSL                   No  ...
7037          Yes                No              No  No internet service  ...
7038          Yes               Yes             DSL                  Yes  ...
7039          Yes               Yes     Fiber optic                   No  ...
7040           No  No phone service             DSL                  Yes  ...
7041          Yes               Yes     Fiber optic                   No  ...
```

```
print("\nColumn Names")
print(df.columns)
```

```
Column Names
Index(['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents',
       'tenure', 'PhoneService', 'MultipleLines', 'InternetService',
       'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport',
       'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling',
       'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn'],
      dtype='str')
```

```
# PRACTISE EXPERIMENT-1 Muthuraj(241801176)
# 3. Explore the Dataset
print("\nDataset Information")
print(df.info())
```

```
Dataset Information
<class 'pandas.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   str
 1   gender            7043 non-null   str
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   str
 4   Dependents        7043 non-null   str
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   str
 7   MultipleLines     7043 non-null   str
 8   InternetService   7043 non-null   str
 9   OnlineSecurity    7043 non-null   str
 10  OnlineBackup      7043 non-null   str
 11  DeviceProtection  7043 non-null   str
 12  TechSupport       7043 non-null   str
 13  StreamingTV       7043 non-null   str
 14  StreamingMovies   7043 non-null   str
 15  Contract          7043 non-null   str
 16  PaperlessBilling  7043 non-null   str
```

```
print("\nSummary Statistics")
print(df.describe(include='all'))
```

```
Summary Statistics
       customerID gender  SeniorCitizen Partner Dependents       tenure  \
count        7043   7043    7043.000000    7043       7043  7043.000000
unique       7043      2            NaN       2          2          NaN
top     7590-VHVEG   Male            NaN      No         No          NaN
freq            1   3555            NaN    3641       4933          NaN
mean          NaN    NaN       0.162147     NaN        NaN    32.371149
std           NaN    NaN       0.368612     NaN        NaN    24.559481
min           NaN    NaN       0.000000     NaN        NaN     0.000000
25%           NaN    NaN       0.000000     NaN        NaN     9.000000
50%           NaN    NaN       0.000000     NaN        NaN    29.000000
75%           NaN    NaN       0.000000     NaN        NaN    55.000000
max           NaN    NaN       1.000000     NaN        NaN    72.000000

       PhoneService MultipleLines InternetService OnlineSecurity  ...  \
count          7043          7043            7043           7043  ...
unique            2             3               3              3  ...
top             Yes            No     Fiber optic             No  ...
freq           6361          3390            3096           3498  ...
mean            NaN           NaN             NaN            NaN  ...
std             NaN           NaN             NaN            NaN  ...
min             NaN           NaN             NaN            NaN  ...
25%             NaN           NaN             NaN            NaN  ...
50%             NaN           NaN             NaN            NaN  ...
75%             NaN           NaN             NaN            NaN  ...
max             NaN           NaN             NaN            NaN  ...

       DeviceProtection TechSupport StreamingTV StreamingMovies  \
count              7043        7043        7043            7043
unique                3           3           3               3
top                  No          No          No              No
freq               3095        3473        2810            2785
mean                NaN         NaN         NaN             NaN
std                 NaN         NaN         NaN             NaN
min                 NaN         NaN         NaN             NaN
25%                 NaN         NaN         NaN             NaN
```

```
print("\nMissing Values")
print(df.isnull().sum())
```

```
Missing Values
customerID          0
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64
```

```
# PRACTISE EXPERIMENT-1 Muthuraj(241801176)
# 4(c). Verify Missing Values
print(df.isnull().sum())
```

```
customerID          0
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64
```

```
print("\nDuplicate Records")
print(df.duplicated().sum())
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")
```

```
Duplicate Records
0
```

```
# PRACTISE EXPERIMENT-1 Muthuraj(241801176)
# 4(a). Remove Duplicate Records
print("\nShape Before Removing Duplicates:", df.shape)
df.drop_duplicates(inplace=True)
print("Shape After Removing Duplicates:", df.shape)
```

```
Shape Before Removing Duplicates: (7043, 21)
Shape After Removing Duplicates: (7043, 21)
```

```
# PRACTISE EXPERIMENT-1 Muthuraj(241801176)
# 4(b). Replace Missing Values
df.fillna(df.median(numeric_only=True), inplace=True)

for col in df.select_dtypes(include=['object', 'str']).columns:
    df[col] = df[col].fillna(df[col].mode()[0])
```

```python
# PRACTISE EXPERIMENT-1 Muthuraj(241801176)
# 5. Encode Categorical Columns
le = LabelEncoder()
categorical_columns = [
    'gender',
    'Partner',
    'Dependents',
    'Contract',
    'PaymentMethod',
    'InternetService',
    'Churn'
]
for col in categorical_columns:
    df[col] = le.fit_transform(df[col])

print("\nEncoded Columns")
print(df[categorical_columns].head())
```

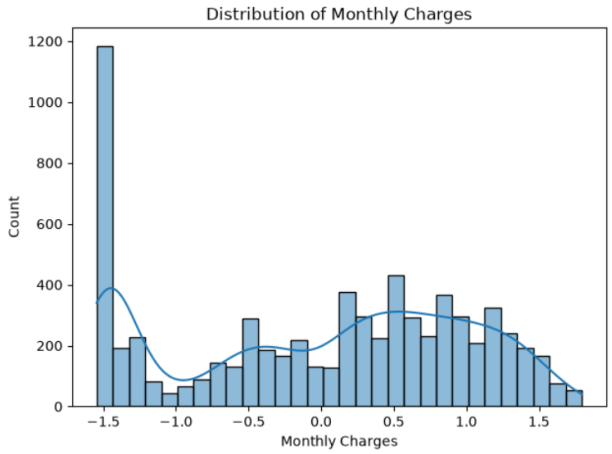
```
Encoded Columns
   gender  Partner  Dependents  Contract  PaymentMethod  InternetService  \
0       0        1           0         0              2                0
1       1        0           0         1              3                0
2       1        0           0         0              3                0
3       1        0           0         1              0                0
4       0        0           0         0              2                1

   Churn
0      0
1      0
2      1
3      0
```

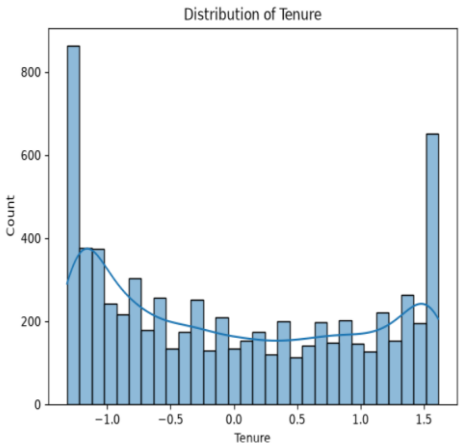
```python
# PRACTISE EXPERIMENT-1 Muthuraj(241801176)
# 6. Standardize Numerical Features
scaler = StandardScaler()
numerical_columns = [
    'tenure',
    'MonthlyCharges',
    'TotalCharges'
]
df[numerical_columns] = scaler.fit_transform(df[numerical_columns])
print("\nScaled Numerical Features")
print(df[numerical_columns].head())
```

```
Scaled Numerical Features
     tenure  MonthlyCharges  TotalCharges
0 -1.277445       -1.160323     -0.994242
1  0.066327       -0.259629     -0.173244
2 -1.236724       -0.362660     -0.959674
3  0.514251       -0.746535     -0.194766
4 -1.236724        0.197365     -0.940470
```

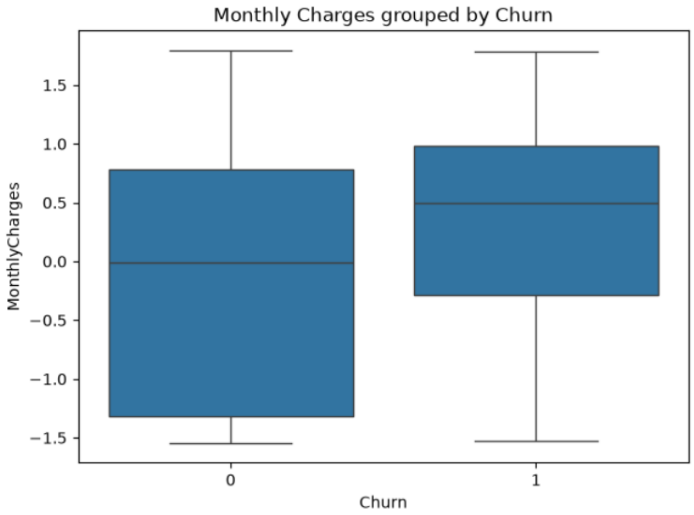
```python
# PRACTISE EXPERIMENT-1 Muthuraj(241801176)
# 7(a). Distribution of MonthlyCharges
plt.figure(figsize=(7,5))
sns.histplot(df['MonthlyCharges'], bins=30, kde=True)
plt.title("Distribution of Monthly Charges")
plt.xlabel("Monthly Charges")
plt.ylabel("Count")
plt.show()
```



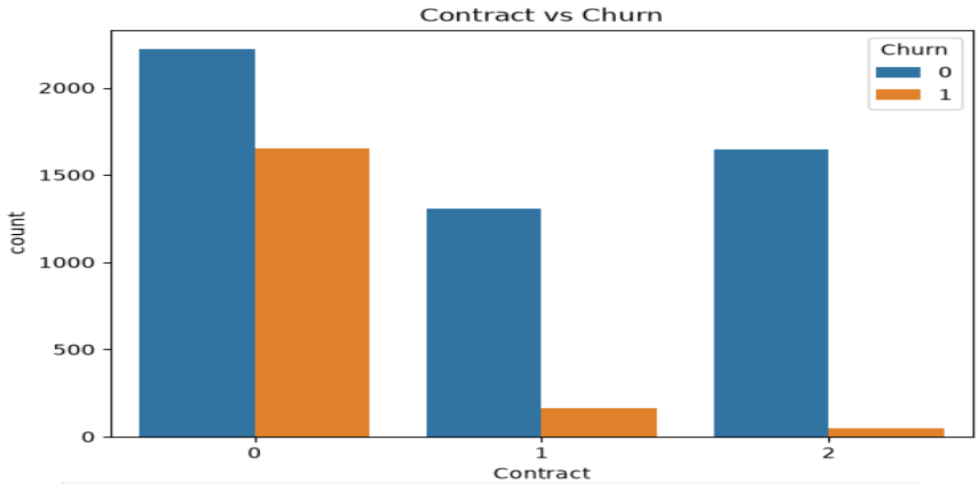

```python
# PRACTISE EXPERIMENT-1 Muthuraj(241801176)
# 7(b). Distribution of Tenure
plt.figure(figsize=(7,5))
sns.histplot(df['tenure'], bins=30, kde=True)
plt.title("Distribution of Tenure")
plt.xlabel("Tenure")
plt.ylabel("Count")
plt.show()
```



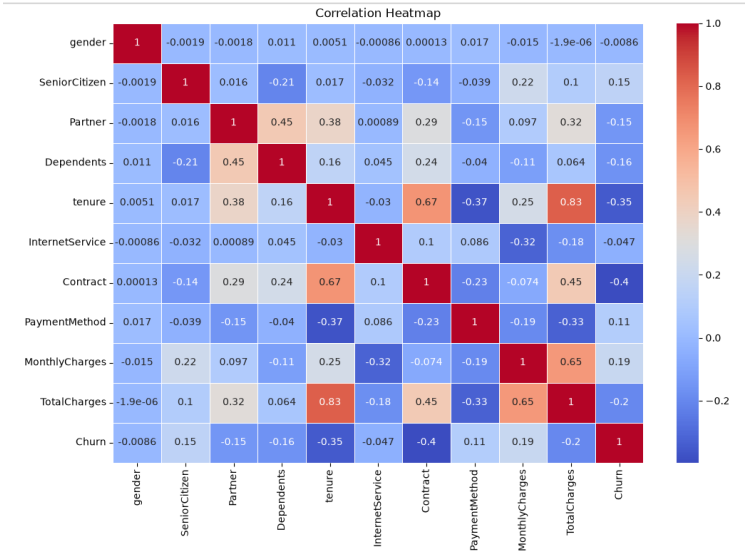
```
# PRACTISE EXPERIMENT-1 Muthuraj(241801176)
# 9. Box Plot
plt.figure(figsize=(7,5))
sns.boxplot(x='Churn', y='MonthlyCharges', data=df)
plt.title("Monthly Charges grouped by Churn")
plt.show()
```




```
# PRACTISE EXPERIMENT-1 Muthuraj(241801176)
# 8. Count Plot
plt.figure(figsize=(7,5))
sns.countplot(x='Contract', hue='Churn', data=df)
plt.title("Contract vs Churn")
plt.show()
```




```
# PRACTISE EXPERIMENT-1 Muthuraj(241801176)
# 10. Correlation Heatmap
corr = df.select_dtypes(include=['int64', 'float64']).corr()
plt.figure(figsize=(12,8))
sns.heatmap(
    corr,
    annot=True,
    cmap="coolwarm",
    linewidths=0.5
)
plt.title("Correlation Heatmap")
plt.show()
```



RESULT:

The Titanic dataset was successfully loaded and analyzed. Dataset information and summary statistics were displayed, missing values in Age, Cabin, and Embarked were handled, and passenger-class distribution was visualized. Passenger records were filtered using the specified conditions. Finally, the dataset was split into 712 training records and 179 testing records.